

SUHAS NAGARAJ

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SKILLS

- **Technical Expertise:** ROS1/ROS2, ML (Supervised and Reinforcement) and DL (Feedforward, CNN and Transformers), NLP & LLMs, RAG, Multimodal and Generative AI, Computer Vision, Simulation, SLAM, 3D reconstruction, Sensor Fusion, Multi-threading & Parallel Programming, CUDA, Robot Modelling and Programming, Human Robot interaction, Cloud Computing, Data structures and Algorithms, Containerization, Version Control, Optimization, Control theory, Manufacturing, Rapid Prototyping, Industrial Robotics, PLC Programming.
- **Programming Languages:** Python, C++, MATLAB, RAPID (ABB), KAREL (FANUC).
- **Libraries, Frameworks and Tools:** PyTorch, Cuda, Tensorflow, Keras, ONNX, Huggingface, Langchain, OpenCV, Open3D, PCL, Yolo, Cartographer, NumPy, pandas, scikit-learn, NLTK, Docker, AWS, Azure, GCP, Linux, GitHub.
- **Others:** MATLAB Simulink, FANUC ROBOGUIDE, ABB Robot Studio, CimStation Robotics, SolidWorks, AutoCAD, Fusion 360, CATIA.

PROFESSIONAL EXPERIENCE

Teaching Assistant [MSML605: Computing Systems for Machine Learning]

January 2025 – Present

College of Computer, Mathematical & Natural Sciences, University of Maryland

College Park, USA

- **Guiding 60 graduate students** in Python fundamentals, machine learning hardware concepts, and efficient use of the university's central HPC cluster (Zaratan) for model training—reducing average computational time by **90%** and enabling more advanced experiments.
- **Collaborating** with Dr. Jerry Wu to revamp course content resulting in a **20% increase** in class participation and engagement.

Research Intern

May 2024 – Present

Intelligent Inclusive Interaction Design (I3D) Lab, Indian Institute of Science (IISc)

Bengaluru, India

- **Developed** a novel 3D reconstruction algorithm integrating LiDAR and Gaussian Splatting SLAM, improving localization accuracy by **200%**.
- **Reduced** computational overhead by **35%**; validated the system in a simulation environment and on a real TurtleBot.
- **Co-authored a paper accepted at IEEE ICRA 2025 (second author, 39% acceptance rate).**

Engineering Intern

October 2022 – March 2023

DiFACTO Robotics and Automation Pvt. Ltd

Bengaluru, India

- **Designed** automation parts using SolidWorks, **performed** industrial process simulations with FANUC RoboGuide across **5 projects**.
- **Programmed ABB (RAPID) and FANUC (KAREL) robots** in both offline and online environments.
- **Developed** PLC programming and DeviceNet interfacing skills to control robot cells using Mitsubishi PLC.

Technical Consultant

August 2021 – March 2022

U-Solar Clean Energy Solutions Pvt. Ltd.

Bengaluru, India

- **Managed** a portfolio of **50+ client accounts** and **10 partner relationships**, contributing to a **20% increase** in customer retention.
- **Spearheaded** market and policy research initiatives, resulting in a **15% growth** in market reach and an expanded project pipeline.
- **Collaborated** with cross-functional teams to devise technical strategies, driving **25% higher** lead generation and overall project success.

PROJECTS (<https://www.suhasnagaraj.com/projects>)

- **Adaptive Text-to-Command Translation for Robotic Navigation:** Fine-tuned T5-Small [w & wo LoRA] LLM to convert natural language commands into structured navigation plans for TurtleBot3 in ROS2 and Gazebo, achieving 100% test accuracy during deployment.
- **ASD Prediction Using Eye Tracking Data:** Developed a novel deep learning approach (CNN, LSTM, Transformers on Pytorch) for predicting (classification) Autism Spectrum Disorder using eye-tracking data, achieving 88.9% accuracy.
- **Automated Guided Vehicle:** Designed autonomous navigation with lane following, stop sign detection, and dynamic obstacle avoidance for mobile robots using ROS2 and OpenCV in simulation and real-world setups, using Turtlebot3 Waffle.
- **Maze Navigation with Object Detection:** Developed ROS2-based system for maze traversal using Aruco markers and object detection
- **SLAM-Based Waypoint Navigation:** Designed an autonomous waypoint navigation system for TurtleBot3 using SLAM, TF tree localization.
- **Deep Reinforcement Learning:** Applied Deep Q-Networks (DQN) for obstacle avoidance and autonomous navigation of mobile robots.
- **MULTI@VIZ:** Developed a ROS2-based GUI for real-time multi-robot coordination, visualization, and control.
- **NLP Projects:** Developed machine learning and NLP models, including feature-enhanced QA buzzers, speech classifiers, poetry checkers, language models, deep neural networks, dependency parsers, and fine-tuned BERT for efficient text and speech processing.
- **Inverse Kinematics & Machine Tending:** Focused on IK-based control of ABB IRB 1600 in Gazebo with ROS2.
- **Path Planning Algorithms:** Developed Dijkstra's, A*, RRT, RRT*, Bidirectional A*, Potential Field and Real Time RRT* algorithms for mobile robot navigation, implemented the solutions on a Turtlebot3 Waffle robot in both simulation and real-world setups.
- **Advanced Control strategies:** Implemented feedback linearization & sliding mode (quadcopter), LQR/LQG (dual-pendulum cart).

EDUCATION

Master of Engineering in Robotics (Artificial Intelligence) [CGPA: 4.0 / 4.0]

August 2023 - May 2025

University Of Maryland, College Park [Phi Kappa Phi Honor Society Member]

College Park, United States

Bachelor of Engineering in Mechanical Engineering [CGPA: 9.3 / 10.0]

August 2017 - July 2021

Ramaiah Institute of Technology

Bengaluru, India

PUBLICATION

Ajay Kumar Sandula, *Shriram Damodaran, ***Suhas Nagaraj**, Debasish Ghose and Pradipta Biswas [* are equally contributing authors]; "3D Reconstruction via Camera-Lidar (2D) Fusion for Mobile Robots: A Gaussian Splatting Approach"; IEEE International Conference on Robotics and Automation (ICRA) **Atlanta 2025**.